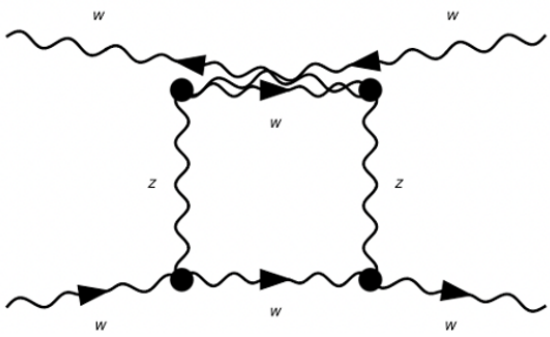
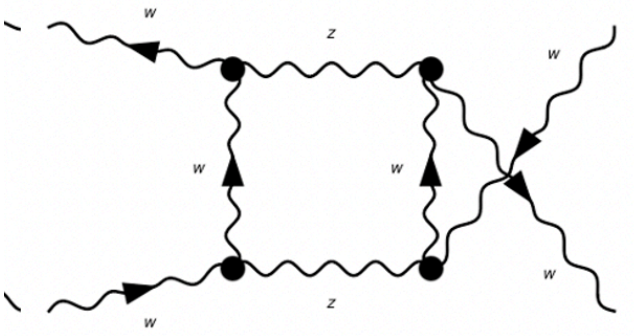
Box-like

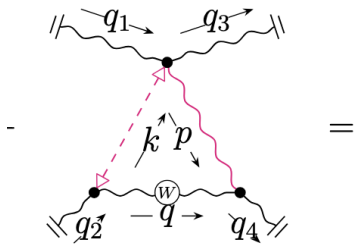
- Make box defaults square if not already. do not want any rectangular boxes
- Implement a cross box diagram that keeps the top and bottom legs normal and crosses the right and left internal legs. I believe we already have a similar topology but where the the left and right are fixed and top and bottom cross
- Also implement cross boxes like this



And this



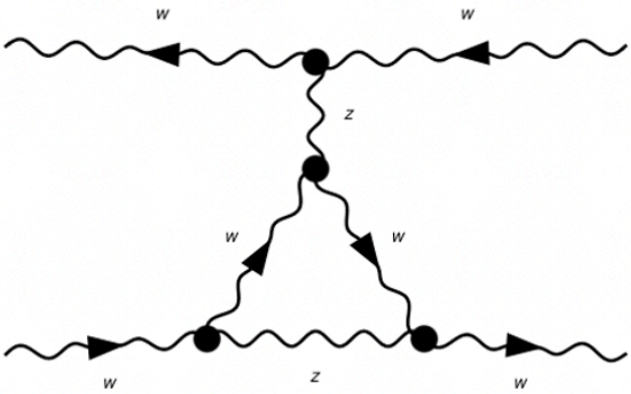
- Implement a 4-legged bubble diagram →
- Completely fix the label position system for this topology (vertex corr?) im not sure what weve been calling it.
- Right now it seems the labels are all wired together. They should all have their own positions and leg positions that are pink in this diagram should not have labels on the inside of the triangle the labels should always be positioned on the outside of the triangle. So, whatever defaults we land on should be like that. Also i dont think the top and bottom momentum labels should be linked in side because their respective edge lengths are not the same and this will prevent everything from being as cohesive as it should be.



- Ensure we can produce all 4 orientations of this diagram because we might want to show all of them in my paper.

Real triangle correction

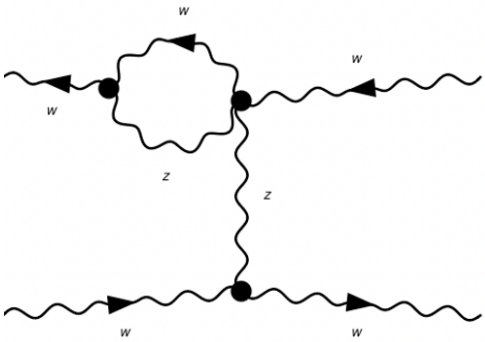
- Implement the regular triangle diagram in all four versions: top and bottom for t channel and left and right for s-channel
- Some examples



T-channel 1 loop corrs

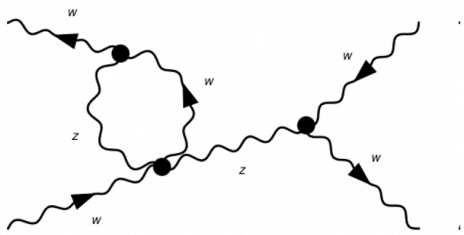
- Seagull and self-energy insertions implementation on all four legs → not just one
- Implement a new swordfish topology
    - Should work for all four legs for t channel and all for legs for s channel. Do not need it to work for multiple legs at once just one leg at a time.

    - The implementation I can already foresee being very similar to the self energy except there being no space between the center vertex the bubble is attached to and the bubble.



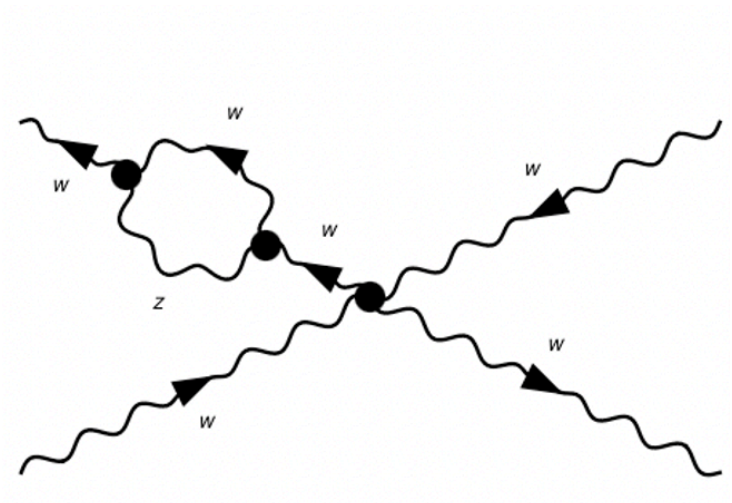
- We also need

S-channel 1 loop corrs

- seagull and self-energy insertions must also work on s-channel trees
- Implement a new swordfish topology like this:
- 

Contact 1 loop corrs

- We already wrote about this earlier but continuing here for the other ones
- Implement the vertex corrs where we can have Z boson leg that goes from q1 to q4 and q3 to q2
- Ours should preferably be more square shaped
- So vertex corrs,



Vertex correction labelling system

For all topologies lets make sure the proca arrows are correctly hitting the vertices

Ensure reference is completely up to day. Possible later rework of it. Also rework of our github landing page, visuals that are there, etc